# ATP Match Performance Analysis & Training Resource Allocation

**To:** Director of Player Development & Coaching Staff
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**Repository:** https://github.com/ryan-salas/ADS-505-Tennis-Analytics

## Executive Summary

Across 22,883 retained ATP matches from 2015–2023, first- and second-serve points won were much more strongly differentiated between winners and losers than first-serve-in percentage. A retrospective logistic-regression audit reached 93.82% validation accuracy in 2021 and 93.13% on held-out 2022–2023 matches, versus a 50.89% majority baseline on validation. Because the classifier uses completed-match statistics, this is not a pre-match forecast and does not establish the causal effect of changing practice. We recommend that the Director of Player Development authorize a time-limited pilot emphasizing serve placement, serve-plus-one execution, and second-serve quality, with coach-led monitoring and a controlled evaluation before changing regular allocations.

## Decision Context

The director allocates scarce court hours. The decision is whether to **pilot** an efficiency-focused adjustment to serve practice and evaluate whether it improves measurable player outcomes. An incorrect “successful performance” classification could cause missed intervention, while an incorrect “poor performance” classification could waste practice time on unnecessary changes. Dollar costs and the exact optimal training allocation are not estimated from the dataset; both require institutional input and prospective testing.

## Evidence from Exploratory Analysis

| Performance indicator | Winners | Losers | Difference |
|---|---|---|---|
| First-serve points won | 77.09% | 66.85% | +10.24 pp |
| Second-serve points won | 56.72% | 45.88% | +10.84 pp |
| Break points saved | 66.85% | 52.01% | +14.84 pp |
| First serves in | 62.07% | 60.27% | +1.80 pp |

Training-period sample: 14,898 matches, 2015–2020. **Takeaway:** Point-winning efficiency is more distinctive of winning than raw first-serve consistency; the association is not evidence that any one drill causes improved results.

## Model Performance and Practical Value

| Evaluation | Majority baseline | Logistic regression | Random forest |
|---|---|---|---|
| 2021 validation accuracy | 50.89% | 93.82% | 93.40% |
| 2021 validation ROC AUC | 0.5000 | 0.9854 | 0.9831 |
| 2022–2023 final test accuracy | — | 93.13% | Not reported |

The models summarize how reliably match-performance rate differences correspond to the recorded winner. This indicates possible value in consistent post-match benchmarking; revenue gains, saved coaching hours, and avoided losses have **not** been demonstrated. Model errors still exist even at high average accuracy. Subgroup validation accuracy for logistic regression was 93.07% on hard, 94.21% on clay, and 96.59% on grass (2021).

## Recommendation and Call to Action

Approve a monitored 6–8 week serve-efficiency training pilot. Introduce targeted practice in (1) first-serve placement and the subsequent shot, (2) second-serve depth and return-pressure scenarios, and (3) high-leverage break-point execution. The original proposal to redirect 40% of serve-drilling time should be treated as a **starting hypothesis**, adjusted for coaching judgment and workload rather than a proven optimal target. Compare participating players against their own baseline and, where feasible, a control group using opponent- and surface-aware metrics. Advance to broader rollout only if gains are credible and practical.

## Confidence, Risks, and Failure Modes

Confidence is relatively high that these retrospective associations recur in the studied ATP main-tour data and across the tested seasons, but low that the data identify a causal training effect or an optimal budget split. Risks include missing-statistics selection bias (1,814 of 24,697 raw records were excluded), weaker representation in some event types, match-length and opponent-strength confounding, and misleading interpretation of end-of-match features as pre-match information. The tool should not be used for standalone player selection, pre-match betting, or automatic practice assignments.

## References and AI-Use Disclosure

migumax. (n.d.). *tennis_atp* [Data set]. GitHub. https://github.com/migumax/tennis_atp
Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830. Full author list is recorded in the technical notebook.

AI assistance supported preliminary code/pipeline ideas, editing, and communication design. The team must verify claims, explain its own modeling decisions, and finalize a fully accurate AI reflection log before submission.